Relational Algebra: Assignment 3 + Quiz 3

Course: COMP 3005AB — Database Management Systems
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1 Problem 1: Telephone Switch RA Specifications

Mathematical logic for queries P1.01–P1.10 based on Switch Model Rev 4.

1. **P1.01 (Directory):** $\Pi_{name,address,area,office,number}(S \bowtie_{S.portid=L.portid} L)$
2. **P1.02 (Office Count):** ${}_{area,office}\mathcal{G}_{COUNT(*)\rightarrow num_sub}(L)$
3. **P1.03 (Intra-switch):** $\Pi_{name}(S \bowtie_{S.portid=C.orig} (C \bowtie_{orig=L1} L1 \bowtie_{term=L2} L2))$
4. **P1.04 (No Origination):** $\Pi_{name}(S) - \Pi_{name}(S \bowtie_{S.portid=C.orig} C)$
5. **P1.05 (Total Inactivity):** $\Pi_{name}(S) - \Pi_{name}(S \bowtie_{S.orig} C) - \Pi_{name}(S \bowtie_{S.term} C)$
6. **P1.06 (Power Users/Division):** $\Pi_{name,service}(S \bowtie SS) \div \Pi_{scode}(V)$
7. **P1.07 (Trunk Load):** ${}_{foreign_sw}\mathcal{G}_{COUNT(call_id)}(T \bowtie_{T.portid=C.orig \vee T.portid=C.term} C)$
8. **P1.08 (Service Rank):** $\tau_{sub_count\downarrow}({}_{service}\mathcal{G}_{COUNT(portid)}(V \bowtie_{left} SS))$
9. **P1.09 (Area 613):** $\Pi_{portid,foreign_sw}(T \bowtie \sigma_{area='613'}(TR))$
10. **P1.10 (Facility State):** $\Pi_{portid}(\sigma_{state='BUSY'}(CH))$

2 Problem 2: myTunes Assembly RA Logic

Formalizing the ETL constraints for the unified database assembly.

11. **Orphan Pruning Logic:**
    Represent the set of all songs in the master list that have a corresponding entry in the CD table (ensuring no orphans exist).

Solution: $ValidSongs = Songs \bowtie_{Songs.cd_id=CDs.cd_id} CDs$

12. **Orphan Identification (The 204 Records):**
How the initial 204 orphans were identified using set difference.

Solution: $Orphans = \Pi_{cd_id}(Songs) - \Pi_{cd_id}(CDs)$

3 Quiz 3 Performance Analysis

Correction for Question 2

The Error: Misidentifying the natural join ($*$) requirement for non-identical column names (*service* vs *scode*).
The RA Fix: $SERVICE_COUNT(scode, n) = (service\mathcal{G}_{count(portid)}SS)$. By renaming the attribute to *scode*, the natural join with the *Services* table becomes valid.

4 Quiz 3 - Relational Algebra Review

Questions based on the Telephone Switching Database Rev 4 schema.

1. (1 point) Which ONE of the following will list the call_id's for all calls that both originate and terminate in a foreign switch (where both originator and terminator are trunks)?

   A. $\pi_{\text{call_id}}(\text{calls} * \text{trunks} * \text{trunks})$

   B. $\pi_{\text{call_id}}(\text{calls} \bowtie_{\text{orig=portid} \wedge \text{term=portid}} \text{trunks})$

   C. $\pi_{\text{call_id}}(\text{calls} \bowtie_{\text{orig=portid} \wedge \text{term=portid}} \text{trunks})$

   D. $\sigma_{\text{call_id}}(\text{calls} \bowtie_{\text{orig=term}} \text{trunks})$

   E. $\pi_{\textbf{call_id}}(\pi_{\textbf{call_id, term}}(\textbf{calls} \bowtie_{\textbf{orig=portid}} \textbf{trunks}) \bowtie_{\textbf{term=portid}} \textbf{trunks})$

2. (1 point) Which ONE of the following would list the names of all the services and the number of lines that subscribe to that service?

   A. SERVICE_COUNT(scode, num_services) =
   $(\text{service}\mathfrak{F}_{\text{count portid}}(\text{service_subscribers}))$
   $\pi_{\text{name, num_services}}(\text{subscribers} * \text{SERVICE_COUNT})$

   B. $\textbf{SERVICE_COUNT(scode, num_services)} =$
   $(\textbf{service}\mathfrak{F}_{\textbf{count portid}}(\textbf{service_subscribers}))$
   $\pi_{\textbf{service, num_services}}(\textbf{services} * \textbf{SERVICE_COUNT})$

C. SERVICE_COUNT(service, num_services) =
(service$\mathfrak{F}_{\text{count portid}}$(service_subscribers))
$\pi_{\text{service, num_services}}$(services $*$ SERVICE_COUNT)

D. $\pi_{\text{service, num_services}}$((service$\mathfrak{F}_{\text{count portid}}$(service_subscribers)))

Solution: Renaming the grouping attribute to scode is necessary because service in service_subscribers is a foreign key, but service in the services table refers to the descriptive name. A natural join ($*$) requires identical column names to link the tables.

3. (1 point) Which ONE of the following would list only the portid's of the trunks that currently have at least one idle channel?

A. $\pi_{\text{state}}(\sigma_{\text{state='IDLE'}}(\text{channels} * \text{trunks}))$

B. $\pi_{\mathbf{portid}}(\mathbf{trunks} * \pi_{\mathbf{portid}}(\sigma_{\mathbf{state='IDLE'}}(\mathbf{channels})))$

C. $\pi_{\text{portid}}(\sigma_{\text{state='IDLE'}}(\text{trunks}))$

D. $\pi_{\text{portid}}(\sigma_{\text{state='IDLE'}}(\text{channels}))$

E. $\pi_{\text{portid}}((\text{calls} \bowtie_{\text{orig=portid}} \text{trunks}) * \pi_{\text{portid}}(\sigma_{\text{state='IDLE'}}(\text{channels})))$

4. (1 point) Suppose a Line originator dialed (905) 238-1243. Which ONE of the following would find the portid's of all the trunks that could route this call (including wildcards)?

A. $\sigma_{\text{portid}}(\sigma_{\text{area='905'} \vee \text{area='000'}}(\text{trunks} * \text{channels}))$

B. $\pi_{\text{portid}}(\sigma_{\text{area='905'} \vee \text{area='000'}}(\text{trunk_routes}))$

C. $\pi_{\mathbf{portid}}(\sigma_{(\mathbf{area='905'} \wedge \mathbf{office='238'}) \vee (\mathbf{area='905'} \wedge \mathbf{office='000'}) \vee (\mathbf{area='000'} \wedge \mathbf{office='000'})}(\mathbf{trunk_routes}))$

D. $\pi_{\text{portid}}(\sigma_{\text{state='IDLE'}}(\text{trunk_routes}))$

E. $\pi_{\text{portid}}(\sigma_{\text{foreign_switch='905,238'}}(\text{trunks}))$

5. (1 point) Which ONE of these relational algebra queries will find the address of everyone that subscribes to the 'Call Waiting' service (scode='CWT')?

A. $\sigma_{\text{address}}(\sigma_{\text{service='Call Waiting'}}(\text{service_subscribers}))$

B. $\sigma_{\text{address}}(\text{subscribers} * \mathfrak{F}_{\text{COUNT scode}}(\text{services}))$

C. $\pi_{\text{address}}(\text{service_subscribers} * \text{subscribers})$

D. $\pi_{\text{address}}(\sigma_{\text{service='CWT'}}(\text{service_subscribers} \bowtie_{\text{service='Call Waiting'}} \text{subscribers}))$

E. $\pi_{\mathbf{address}}(\sigma_{\mathbf{service='CWT'}}(\mathbf{service_subscribers} * \mathbf{subscribers}))$